Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER ON-LINE PROCTORED EXAMINATIONS

SOLID STATE DRIVES [ELE 4088]

REVISED CREDIT SYSTEM

Time: 75 Minutes + 10 Minutes **Date: 17 December 2021** **Max. Marks: 20**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Time: 75 minutes for writing + 10 minutes for uploading.

1A. The load inertia referred to the motor shaft of a conveyor belt system is 0.05kgm^2. The materials are transported at the rate of 3600Kg/hr. The belt is of 100 meter long and moves at a uniform speed of 2 m/s. Determine the speed at which the motor is driven? ***(05)***

1B. Speed torque curve of a motor is shown in Figure 1B. Draw load curves which will give stable operation with the portion of characteristics marked as **AB**. Justify your answer.

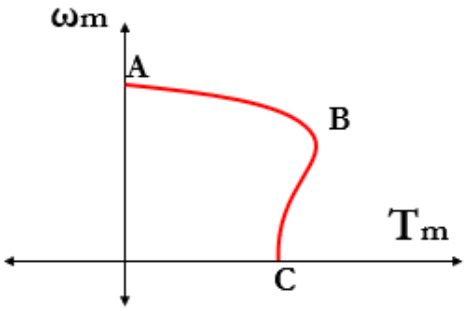


Figure 1B. ***(03)***

1C. Differentiate between true synchronous mode and self-control mode for variable frequency control of Synchronous motor. Suggest which method is suitable for pump drive. ***(02)***

2A. A 220V, 1500 rpm, 12 A separately excited DC motor has armature resistance and inductance of 2Ω and 30mH respectively. This motor is controlled by a 1-phase fully controlled rectifier with an AC source of 230V, 50Hz. Identify the mode and calculate the speed for α= 30° and T =30Nm. ***(04)***

2B. With a help of a block diagram show how closed loop speed control can be implemented in a single-phase controlled rectifier fed separately excited DC motor. ***(03)***

2C. A 440V, 3 phase, 50Hz, 6 pole, 945 rpm, delta connected induction motor has the rotor resistance referred to the stator $Rr^1=2\Omega$. Neglect the stator impedance and rotor leakage reactance.

The motor speed is controlled by stator voltage control. The motor drives a load whose torque varies linearly with speed i.e $T_L=k(1-s)$.

Determine the motor speed and torque when the terminal voltage is 280V. ***(03)***